

AI Advisor Response

Question:

explain concepts of oops

Answer:

Object-oriented programming (OOP) organizes code using classes and objects. A class defines data and behavior, while an object is a real instance of the class.

****Key Concepts:****

* ****Purpose:**** It is useful for large projects because it groups related logic, improves code reuse, and makes systems easier to extend.

* ****Design Principles:**** Important principles include single responsibility (a class should have one main reason to change), open-closed, dependency inversion, low coupling, and high cohesion.

****Example:****

```
```python
class Pipeline:
 pass
```
```

****Common Mistakes:****

* Do not use OOP just to look advanced. If a function is enough, use a function.

* Use OOP specifically when state and behavior belong together.

****Interview Focus:****

When discussing OOP, explain it as a way to model real components and manage complexity. Be prepared to provide a real project use case and explain the principles listed above.